

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1550

Wards Hollow 138 kV
September 19, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1550. The Interconnection Customer proposes a 209.0 MW battery energy storage facility interconnecting at the Wards Hollow 138 kV substation in the West Virginia area. The request was received on August 19, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 561 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$42,944,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Poplar Bluff Generating Sta / Westone Sw Yd 138.0 kV	Hubbardstown Substation / Tristate 345.0 kV	135.5	280	6.9
Poplar Bluff Generating Station 345-138 kV Transformer Bank	Tristate / Hubbardstown 345.0 kV	123.6	418	9.0
Valemout 138-1 kV Auto	Pence / Valemout 345.0 kV L	121.0	448	5.7
Chemical / Alum Creek 138.0 kV L	Tristate / Valmount 345.0 kV	118.6	316	17.4
Wright Tap / Menaul 138.0 kV Line	Clinton Atle / Lodi Stig Substation 138.0 kV L	110.7	223	100.0
Wright Tap / Peytona 138.0 kV	Lodi Stig / Clinton Atle 138.0 kV Line	110.5	226	100.0
Peytona / San Xavier 138.0 kV	Clinton Atle / Lodi Stig 138.0 kV	109.7	230	100.0
West Sacramento / San Jose Sta. A 138.0 kV	Farragut / Wards Hollow 138.0 kV	104.8	215	34.0
Huntington Jct / Westone Sw Yd 138.0 kV	Tristate / Valemout 345.0 kV Line	104.4	267	9.9
Huntington Junction / Alum Crk 138.0 kV	Valemout / Tristate 345.0 kV line	104.3	264	9.9
Lexington Health Ctr / MattoonE 138.0 kV	Coulee Ave Substation / Amos 345.0 kV	104.2	216	6.1
Dixieland Prison / Lexington Health Ctr 138.0 kV	Coulee Ave / Amos 345.0 kV Line	103.6	221	6.1
Dixieland Prison / New Gulf 138.0 kV	Coulee Ave / Amos 345.0 kV	102.9	225	6.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Poplar Bluff Generating Station / Westone Sw Yd 138.0 kV Line	Rebuild line with higher-capacity conductor	\$9,677,000	10% (\$972,000)
Poplar Bluff Generating Station 345-138 kV Transformer Bank	Replace transformer with higher-rated unit	\$32,522,000	7% (\$2,406,000)
Valemount 138-1 kV Auto	Replace transformer with higher-rated unit	\$6,351,000	5% (\$343,000)
Chemical / Alum Crk 138.0 kV	Rebuild line with higher-capacity conductor	\$8,530,000	62% (\$5,279,000)
Wright Tap / Menaul 138.0 kV Line	Rebuild line with higher-capacity conductor	\$7,421,000	100% (\$7,421,000)
Wright Tap / Peytona 138.0 kV	Rebuild line with higher-capacity conductor	\$11,417,000	100% (\$11,417,000)
San Xavier / Peytona 138.0 kV	Reconductor line	\$4,225,000	100% (\$4,225,000)
West Sacramento / San Jose Sta. A 138.0 kV	Reconductor line	\$2,583,000	42% (\$1,092,000)
Huntington Jct / Westone Sw Yd 138.0 kV	Reconductor line	\$2,591,000	66% (\$1,719,000)
Huntington Junction / Alum Crk 138.0 kV	Reconductor line	\$6,903,000	66% (\$4,581,000)
Lexington Health Ctr / MattoonE 138.0 kV	Reconductor line	\$2,670,000	45% (\$1,193,000)
Dixieland Prison / Lexington Health Ctr 138.0 kV	Reconductor line	\$3,069,000	45% (\$1,372,000)
Dixieland Prison / New Gulf 138.0 kV	Reconductor line	\$2,068,000	45% (\$924,000)
	Total allocated to project		\$42,944,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.